

WEEK 3 REPORT

Superstore Business Analysis Using Power BI

Introduction.

Power BI is a useful tool for converting raw data into meaningful information. In this project, the Superstore dataset was analyzed using Power BI.

The analysis focuses on sales, profit, customers, products, quantity and regions. Different Power BI features such as Power Query, DAX, charts, slicers and KPI cards were used to understand the data.

Objectives.

The main objectives are:

- *To load the Superstore dataset into Power BI.
- *To prepare and clean the data.
- *To create a useful Date Table.
- *To organize dates using a hierarchy.
- *To connect related tables.

*To calculate important business values using DAX.

*To study sales and profit trends.

*To design an interactive dashboard.

Data Used.

The Superstore dataset contains information about orders, customers, products and business performance.

The major fields used in the analysis are:

*Order Date.

*Ship Date.

*Customer ID.

*Customer Name.

*Product ID.

*Product Name.

*Category.

*Sub-Category.

*Segment.

- *Region.

- *Sales.

- *Quantity.

- *Profit.

The data was also checked for missing values, duplicate records and incorret data types before analysis.

Data Preparation.

Power Query was used to prepare the dataset before creating the dashboard.

The following tasks were performed:

- *Checked column names.

- *Verified data types.

- *Checked missing values.

- *Checked duplicate records.

- *Converted date columns to Date format.

- *Converted sales and profit to Decimal Number.

- *Converted quantity to Whole Number.

- *Set category and region fields as Text.

Creating Date Table.

A separate Date Table was created using DAX.

The Date Table contains:

*Date

*Year

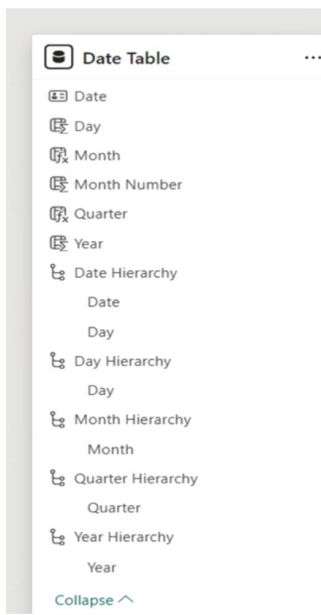
*Quarter

*Month

*Month Number

*Day

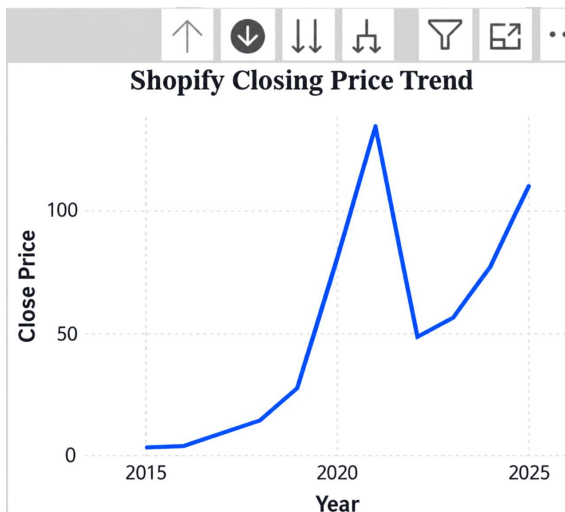
*Day of Week.



Date Hierarchy.

A date hierarchy was created for easy drill-down analysis.

Year → Quarter → Month → Day



This allows the user to view data from yearly level and then move down to quarter, month and day level.

Table Relationship.

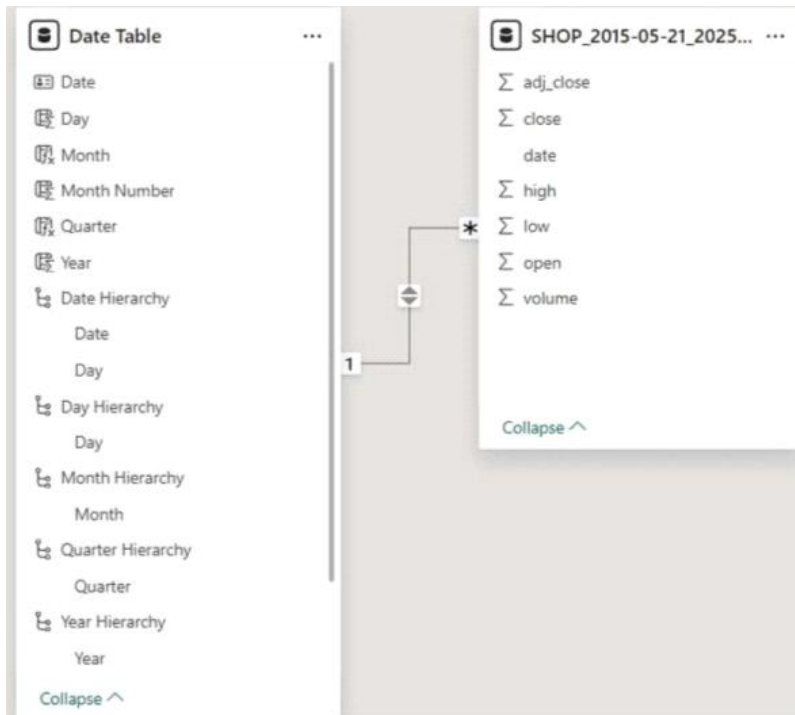
A relationship was created between the Date Table and Superstore Sales table.

Relationship:

*Date Table[Date] → Superstore Sales[Order Date]

Relationship Type: One-to-Many (1:)

*This relationship allows the Date Table to filter the sales data and supports time-based analysis.



DAX Measures.

DAX measures were created to calculate important business values.

Total Sales.

Total Sales = SUM(Superstore[Sales])

Total Profit.

Total Profit = SUM(Superstore[Profit])

Total Quantity.

Total Quantity = SUM(Superstore[Quantity])

Total Orders.

Total Orders = DISTINCTCOUNT(Superstore[Order ID])

Average Sales.

Average Sales = AVERAGE(Superstore[Sales])

Profit Margin.

Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)

These measures help in understanding sales, profit, quantity and order performance.

Time Analysis.

Time-based calculations were used to compare business performance.

The functions included:

***DATEADD.**

***TOTALYTD.**

***SAMEPERIODLASTYEAR.**

These functions help to understand changes in sales and profit over time.

Dashboard Design.

A Power BI dashboard was created to display the results in an easy - to - understand format.

The dashboard contains KPI cards for:

*Total Sales.

*Total Profit.

*Total Orders.

*Total Quantity.

*Average Sales.

*Profit Margin %.

Slicers were added for:

*Year

*Category

*Region

*Segment

Visuals included:

*Sales by Category

*Profit by Region

*Sales Trend

*Sales and Profit Analysis

*Decomposition Tree

These visuals allow users to interact with the report and explore the data.

Analysis.

The dashboard can be used to understand different aspects of the Superstore business.

The analysis helps to:

- 1.View overall sales performance.
- 2.Understand profit performance.
- 3.Compare different product categories.
- 4.Study regional performance.
- 5.Observe sales changes over time.
- 6.Compare customer segments.
- 7.Filter the report using slicers.

8.Explore sales using the decomposition tree.

The interactive visuals make the analysis easier to understand.

Conclusion.

This project provided practical experience in using Power BI for business data analysis. The Superstore dataset was prepared using Power Query and organized using a Date Table and relationships.

DAX measures were created to calculate important business metrics. Time-based analysis was also performed using date functions.

Finally, an interactive dashboard was created using KPI cards, charts, slicers and a decomposition tree. This project helped in understanding the basic process of transforming raw data into useful business insights.

